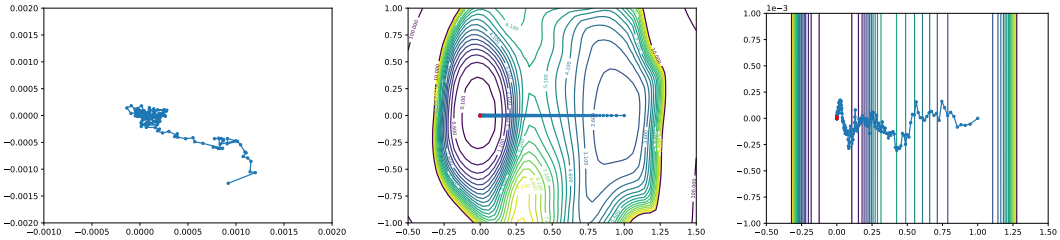


(a) Two random directions (b) Random direction for y-axis (c) Enlarged version (b)

Figure 8: Ineffective visualizations of optimizer trajectories. These visualizations suffer from the orthogonality of random directions in high dimensions.

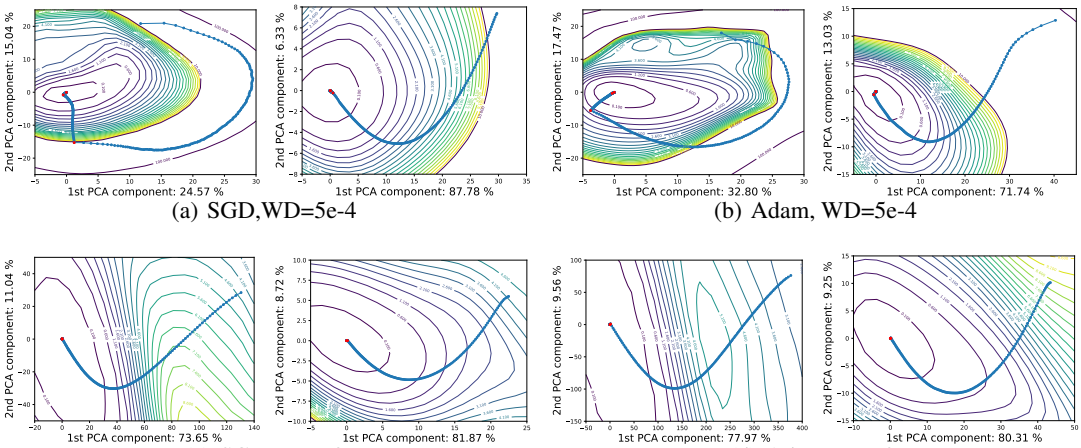
This is problematic when optimization trajectories lie in extremely low dimensional spaces. In this case, a randomly chosen vector will lie orthogonal to the low-rank space containing the optimization path, and a projection onto a random direction will capture almost no variation. Figure 8(b) suggests that optimization trajectories are low-dimensional because the random direction captures orders of magnitude less variation than the vector that points along the optimization path. Below, we use PCA directions to directly validate this low dimensionality, and also to produce effective visualizations.

7.2 Effective Trajectory Plotting using PCA Directions

To capture variation in trajectories, we need to use non-random (and carefully chosen) directions. Here, we suggest an approach based on PCA that allows us to measure how much variation we've captured; we also provide plots of these trajectories along the contours of the loss surface.

Let θ_i denote model parameters at epoch i, and the final parameters after n epochs of training are denoted θ_n. Given n training epochs, we can apply PCA to the matrix $M = [\theta_0 - \theta_n; \cdots; \theta_{n-1} - \theta_n]$, and then select the two most explanatory directions. Optimizer trajectories (blue dots) and loss surfaces along PCA directions are shown in Figure 9. Epochs where the learning rate was decreased are shown as red dots. On each axis, we measure the amount of variation in the descent path captured by that PCA direction.

At early stages of training, the paths tend to move perpendicular to the contours of the loss surface, i.e., along the gradient directions as one would expect from non-stochastic gradient descent. The stochasticity becomes fairly pronounced in several plots during the later stages of training. This is particularly true of the plots that use weight decay and small batches (which leads to more gradient



(a) SGD,WD=5e-4 (b) Adam, WD=5e-4

(c) SGD, WD=0 (d) Adam,WD=0

Figure 9: Projected learning trajectories use normalized PCA directions for VGG-9. The left plot in each subfigure uses batch size 128, and the right one uses batch size 8192.

noise, and a more radical departure from deterministic gradient directions). When weight decay and small batches are used, we see the path turn nearly parallel to the contours and "orbit" the solution when the stepsize is large. When the stepsize is dropped (at the red dot), the effective noise in the system decreases, and we see a kink in the path as the trajectory falls into the nearest local minimizer.

Finally, we can directly observe that the descent path is very low dimensional: between 40% and 90% of the variation in the descent paths lies in a space of only 2 dimensions. The optimization trajectories in Figure 9 appear to be dominated by movement in the direction of a nearby attractor. This low dimensionality is compatible with the observations in Section 6, where we observed that non-chaotic landscapes are dominated by wide, nearly convex minimizers.

8 Conclusion

We presented a visualization technique that provides insights into the consequences of a variety of choices facing the neural network practitioner, including network architecture, optimizer selection, and batch size. Neural networks have advanced dramatically in recent years, largely on the back of anecdotal knowledge and theoretical results with complex assumptions. For progress to continue to be made, a more general understanding of the structure of neural networks is needed. Our hope is that effective visualization, when coupled with continued advances in theory, can result in faster training, simpler models, and better generalization.

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